# Vidya Sagar Reddy Venna

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## EDUCATION

**University of Southern California, Viterbi School of Engineering** — Los Angeles, CA
*Master of Science, Electrical and Computer Engineering (Machine Learning and Data Science)* — Aug 2024 - May 2026
**Relevant Coursework:** Data Structures and Algorithms, Deep Learning, Machine Learning, Image Processing, Robotic Mobility

**Indian Institute of Information Technology, Sri City** — Sri City, India
*Bachelor of Technology (Honours), Electronics and Communication Engineering* — Aug 2020 - May 2024
**Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Embedded Systems, Signal Processing

## EXPERIENCE

**USC Information Sciences Institute, Project Alexandria** — Los Angeles, CA
*Graduate Researcher* — Aug 2025 - Present

- Built a PDF to structured record service in Python, 2,910 lines across 8 modules, with cloud and fully local model backends interchangeable behind one interface, cutting extraction cost per  page paper and to zero on the local path.
- Implemented **semantic retrieval** with ChromaDB and sentence transformer embeddings plus a Neo4j layer that resolves repeated entities across documents via MERGE on (name, type), producing a node,  edge cross document graph instead of per file silos.
- Built a four family automated evaluation harness covering information retention, n-gram overlap, LLM as judge scoring and graph topology, instrumented with Weights and Biases so every run emits reproducible quality signals.

**USC Clinical Biomechanics, Orthopedic and Sports Outcomes Research Lab** — Los Angeles, CA
*Graduate Student Researcher* — Jul 2025 - Present

- Engineered a Python signal processing pipeline for **multi channel** surface EMG from an NIH funded trial, 2 kHz across 7 muscles and 54 patients: notch and Butterworth filtering, TKEO onset detection and MVIC normalization into a QC gated dataset.
- Applied linear mixed effects models with Bonferroni correction and leave one out cross validated logistic regression to show force and neuromuscular timing recover independently ($r = -0.012$, $p = 0.938$), a result carried by two APTA CSM abstracts and a manuscript.

**BrainSightAI** — Bangalore, India
*Machine Learning Intern* — Jan 2024 - Jun 2024

- Processed clinical 3T patient MRI scans using **UNet3D** deep learning models (VoxelBox) to extract DTI metrics and 3D white matter tractography and conducting visual QC on FA maps to eliminate spatial artifacts and false fiber streamlines
- Caught a denoising failure that aggregate SNR and RMSE passed, blocky spatial artifacts visible only on rendered FA maps, and changed the production default on that evidence.

**Human Movement Analysis Lab, IIIT Sri City** — Sri City, India
*Honours Student Researcher* — May 2022 - May 2024

- Led collection and public release of four surface EMG datasets (**EMAHA-DB4 to DB7**, 70 subjects) and trained CNN and BiLSTM classifiers over a filter, rectify, normalize and feature extract pipeline, producing three peer reviewed IEEE publications (EMBC, BIOSIGNALS, ISPA).

## PROJECTS

**Paper Copilot | Python, FastAPI, Docker, PDF.js, GitHub Actions** — 2026

- Built a cited retrieval system over research PDFs with TF-IDF, dense and hybrid Reciprocal Rank Fusion retrievers, grounded page level citations and a refuse on weak evidence path, served through FastAPI with a PDF.js reader, containerized with CI.
- Measured retrieval and generation quality with bootstrap confidence intervals and paired McNemar tests, and documented a negative result where agentic query reformulation degraded answers, shipping it opt in rather than on by default.

**Predictive Maintenance Lab | Python, XGBoost, MLflow, FastAPI, Docker, Azure** — 2026

- Shipped an end to end ML service to Azure Container Apps: training pipeline into an MLflow registry into a FastAPI inference API with Pydantic validation and per route rate limiting, packaged in a multi stage Dockerfile with a compose stack.
- Built GitHub Actions CI gating lint, tests and a container smoke test in sequence plus a scheduled weekly drift report, and benchmarked XGBoost against an LSTM on NASA CMAPSS (RMSE 16.80 vs 22.43) under a leakage safe split.

**Quadruped Locomotion Controller, EE579 | C++, Arduino, I2C** — 2025

- Wrote embedded C++ firmware driving 12 servos through a Buehler clock gait oscillator, with edge triggered landing detection and a software I2C workaround for a driver address conflict, and reported the negative result of the vibration assist test honestly.

## SKILLS

**Languages:** Python, C++, JavaScript, MATLAB, R, SQL
**Software and Systems:** Git, Linux, Docker, FastAPI, GitHub Actions CI, pytest, Azure, spark, GCP, Spark, MLFLow
**AI and Machine Learning:** Pytorch, Tensorflow, Huggingface, Langchain, ChromaDB, Neo4j, PyTorch, scikit-learn, XGBoost, JAX, Weights and Biases, Ollama local inference, ONNX, Time Series, Hybrid Retrival, Opencv

## PUBLICATIONS

**V. S. R. Venna et al., "Impact of Activity Pace and Arm Position on ADL Classification," IEEE EMBC 2024, Florida [Link]**
**V. S. R. Venna et al., "Fine-ADL Classification Using sEMG under Varying Conditions, " BIOSIGNALS 2024, Rome [Link]**
**V. S. R. Venna et al.,"Impact of Measurement Conditions on ADL Classification Using Surface EMG," IEEE ISPA 2023, Rome [Link]**